Skeletal Radiol
DOI 10.1007/s00256-014-2075-3

TECHNICAL REPORT

Percutaneous CT-guided cryoablation for the treatment of refractory pudendal neuralgia

J. David Prologo · Roger C. Lin · Roger Williams · David Corn

Received: 2 September 2014 / Revised: 25 November 2014 / Accepted: 26 November 2014
© ISS 2014

Abstract
Purpose To evaluate the safety and efficacy of percutaneous CT-guided cryoablation of the pudendal nerve for the treatment of refractory pudendal neuralgia.
Materials and methods Eleven patients were selected to undergo percutaneous CT-guided cryoablation of the pudendal nerve based on established diagnostic criteria. Brief Pain Inventory questionnaires were administered prior to the procedure, during the immediate 24 h post procedure, and 45 days and 6 months following the procedure.
Results Prior to treatment, the average level of pain on a scale from 0 (no pain) to 10 (worst pain imaginable) was 7.6, with pain described as "burning" (80%), "pulling" (37.5%), "crushing" (50%), "pressure" (84.5%), "throbbing" (50%), "knife-life" (52%), and "other" (60%). At 24 h, 45 days, and 6 months post-treatment, pain intensity dropped to 2.6, 3.5, and 3.1, respectively ($p<0.005$). There were no procedure-related complications.
Conclusions CT-guided percutaneous cryoablation may represent a safe and efficacious option for selected patients with refractory pudendal neuralgia.

Keywords Pudendal · Nerve · Cryoablation

J. D. Prologo (✉) · R. Williams
Department of Radiology and Imaging Sciences Division of Interventional Radiology and Image-Guided Medicine, Emory University Hospital, 1364 Clifton Road, NE Suite D112, Atlanta, GA 30322, USA
e-mail: john.david.prologo@emory.edu

R. Williams
e-mail: Rwilli3@emory.edu

R. C. Lin
Department of Radiology, University Hospitals Case Medical Center, 11100 Euclid Avenue, Cleveland, OH 44106, USA
e-mail: rogerlin@rogerlin.com

D. Corn
Department of Biomedical Engineering, Case Western Reserve University, 10900 Euclid Avenue, Cleveland, OH 44106-7235, USA
e-mail: djc@case.edu

Published online: 17 December 2014

Introduction

Pudendal neuralgia is a painful condition related to compression, trauma (including gynaecological surgery or traumatic obstetric events), or neuropathy involving the pudendal nerve. Symptoms correspond with the pudendal nerve sensory distribution and are often debilitating [1–6].

Historically treatment options for pudendal neuralgia have included conservative management, physical therapy, surgical intervention, and/or perineural injections of steroids [2, 7–12]. More recently, advanced imaging and nerve stimulator guidance have been employed in attempts to improve outcomes in patients undergoing therapeutic injections [13–19].

The purpose of this study is to evaluate the safety and efficacy of percutaneous CT-guided cryoablation of the pudendal nerve for the treatment of refractory pudendal neuralgia. Strict clinical guidelines—including selection criteria, symptom documentation, and follow-up—were in place for patients who presented with refractory pudendal neuralgia and underwent cryoablation. Data were retrospectively anonymised and analysed for this report.

Methods

Patients

This review was conducted under the principles of the Declaration of Helsinski. Specific strict attention was paid to the

Table 1 Patient demographics

Patient	Duration of Pain (years)	Cause	Pre-procedure pain score. Range (average)	24-H post-procedure pain score. Range (average)	45-Day post-procedure pain score. Range (average)	6-Month post-procedure pain score. Range (average)
1	2.5	Childbirth	4–7 (5)	1–2 (6)	2–2 (2)	0–2 (1)
2	10	Childbirth	7–10 (8)	2–2 (2)	0–3 (2)	0–3 (2)
3	3	Gynaecological surgery	7–7 (7)	0–0 (0)	0–1 (0)	0–3 (2)
4	4	Rectocele repair	8–10 (10)	5–7 (5)	6–6 (6)	0–3 (2)
5	5	Childbirth	3–8 (6)	0–2 (1)	0–2 (1)	0–2 (1)
6	8	Childbirth	8–10 (9)	3–4 (3)	1–3 (3)	0–3 (2)
7	4	Trauma	3–7 (6)	2–3 (2)	0–4 (1)	1–4 (3)
8	5	TURP/radical prostatectomy	8–8 (8)	0–0 (0)	2–2–(2)	0–2–(1)
9	3	Pelvic surgery	8–10 (9)	3–4 (3)	8–10 (9)	8–10 (9)
10	4	Pelvic surgery	8–10 (9)	4–5 (5)	8–10 (9)	8–10 (9)
11	7	Trauma	5–9 (7)	0–2 (2)	1–3 (3)	0–3 (2)

respect for persons, justice, and beneficence principles of the Belmont Report. A total of 11 patients were included (Table 1). To be considered for cryoablation, the patient must have failed conventional treatment for pudendal neuralgia and fulfilled the five main diagnostic criteria of Nantes [1], including CT-guided percutaneous injection of triamcinolone and bupivacaine by the interventionalist planning the cryoablation, with documented positive response.

Procedure

Patients were placed prone on the CT table. Vital signs were continuously monitored throughout the procedure. Pain control and conscious sedation were achieved with intravenous midazolam (0.5–1.5 mg) and fentanyl (50–150 μg). Local anaesthesia with 1% lidocaine was given. A single 17-gauge cryoablation probe (Ice Sphere; Galil Medical, Arden Hills, MN) was advanced to the distal portion of the pudendal canal via a transgluteal approach under CT guidance (Fig. 1). Two 8-min–5-min freeze-thaw intervals were undertaken. The probe was then removed and a bandage placed on the insertion site. The patient was monitored for 2 h after the procedure to assess for any post-procedure complications.

Outcome variables

The Brief Pain Inventory (BPI) was used to measure patient pain prior to and after the procedure (Fig. 2). Follow-up was documented at 24 h, 45 days, and 6 months after the procedure.

Statistical analysis

Data were analysed using the paired samples *t*-test and confirmatory non-parametric evaluations were carried out using Wilcoxon's signed rank test. All statistical analyses were performed using IBM SPSS Statistics (IBM Corp., Armonk, NY). All statistical tests were two-sided and a p-value of less than 0.05 was considered significant.

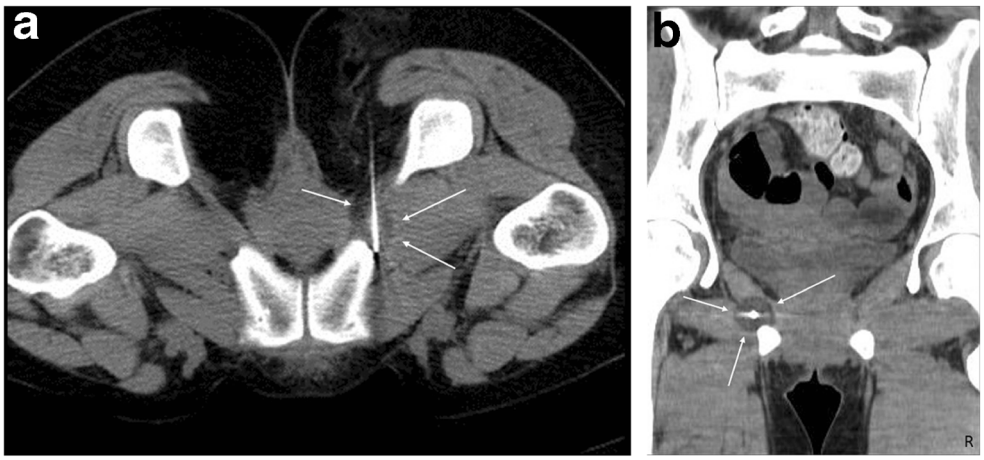


Fig. 1 Axial (**a**) and corresponding coronal (**b**) images of CT-guided cryoablation of a left pudendal nerve in a patient with refractory pudendal neuralgia, distally in the pudendal canal. *Arrows* indicate the periphery of the visualised ablation zone

1903

Date: ☐☐ / ☐☐ / ☐☐
(month) (day) (year)

Subject's Initials : _____

Study Subject #: ☐☐☐☐

PLEASE USE BLACK INK PEN

Study Name: _____

Protocol #: _____

PI: _____

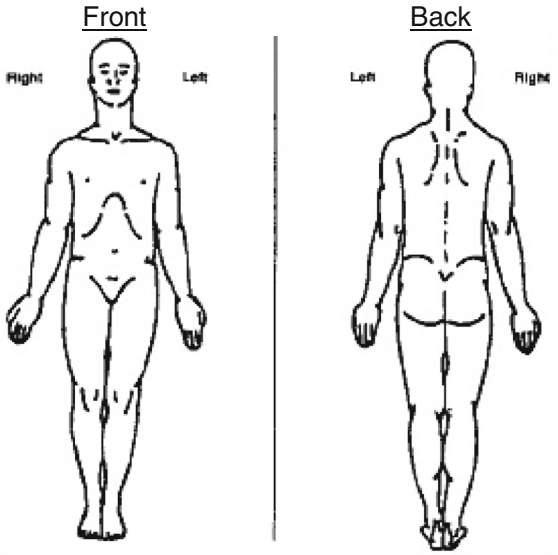
Revision: 07/01/05

Brief Pain Inventory (Short Form)

1. Throughout our lives, most of us have had pain from time to time (such as minor headaches, sprains, and toothaches). Have you had pain other than these everyday kinds of pain today?

☐ Yes ☐ No

2. On the diagram, shade in the areas where you feel pain. Put an X on the area that hurts the most.



3. Please rate your pain by marking the box beside the number that best describes your pain at its worst in the last 24 hours.

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

No Pain — Pain As Bad As You Can Imagine

4. Please rate your pain by marking the box beside the number that best describes your pain at its least in the last 24 hours.

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

No Pain — Pain As Bad As You Can Imagine

5. Please rate your pain by marking the box beside the number that best describes your pain on the average.

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

No Pain — Pain As Bad As You Can Imagine

6. Please rate your pain by marking the box beside the number that tells how much pain you have right now.

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

No Pain — Pain As Bad As You Can Imagine

Copyright 1991 Charles S. Cleeland, PhD
Pain Research Group
All rights reserved

Fig. 2 Brief pain inventory questions

1903

Date: (month) / (day) / (year)

Subject's Initials : _____

Study Subject #:

PLEASE USE BLACK INK PEN

Study Name: _____

Protocol #: _____

PI: _____

Revision: 07/01/05

7. What treatments or medications are you receiving for your pain?

8. In the last 24 hours, how much relief have pain treatments or medications provided? Please mark the box below the percentage that most shows how much relief you have received.

0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
☐ No Relief	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐ Complete Relief

9. Mark the box beside the number that describes how, during the past 24 hours, pain has interfered with your:

A. General Activity

☐ 0 Does Not Interfere ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 Completely Interferes

B. Mood

☐ 0 Does Not Interfere ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 Completely Interferes

C. Walking ability

☐ 0 Does Not Interfere ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 Completely Interferes

D. Normal Work (includes both work outside the home and housework)

☐ 0 Does Not Interfere ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 Completely Interferes

E. Relations with other people

☐ 0 Does Not Interfere ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 Completely Interferes

F. Sleep

☐ 0 Does Not Interfere ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 Completely Interferes

G. Enjoyment of life

☐ 0 Does Not Interfere ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 Completely Interferes

Copyright 1991 Charles S. Cleeland, PhD
Pain Research Group
All rights reserved

Fig. 2 (continued)

Results

A total of 11 patients were included. All patients were followed up at 24 h and 45 days without variation. The 6-month follow-up average corresponded to 204±56 days.

Prior to treatment, the average level of pain on a scale from 0 (no pain) to 10 (worst pain imaginable) was 7.6, with pain described as "burning" (80%), "pulling" (37.5%), "crushing" (50%), "pressure" (84.5%), "throbbing" (50%), "knife-life" (52%), and "other" (60%). At 24 h, 45 days, and 6 months post-treatment, pain intensity dropped to 2.6, 3.5, and 3.1, respectively ($p<0.005$). There were no procedure-related complications.

Two patients, both with predominately perirectal pain, continued to suffer unchanged pain at the 45-day follow-up. Both patients underwent a second, more proximal ablation in the pudendal canal months following the initial ablation [162 and 190 days post-procedure (Fig. 3)]. Both reported an increase in perineal "numbness" following the second procedure but unchanged perirectal pain.

Discussion

Anatomy

The pudendal nerve arises from the S2–S4 nerve roots and exits from the lesser sciatic foramen. It originates as a single trunk in 75% of patients, 22.5% of the cases as two trunks, and exists as three nerve roots in 2.5% of cases [20]. The nerve passes around the ischial spine and then courses along the pudendal canal. In the perineum, the nerve splits into the inferior rectal nerve, perineal nerve, and dorsal nerve of the penis or clitoris [7, 21, 22].

Pudendal neuralgia diagnosis and treatments

The differential diagnosis for pudendal neuralgia includes other causes of pelvic pain such as genitofemoral neuropathy, bulbospongiosus spasm, sacral radiculopathies, and/or piriformis syndrome [23]. Diagnostic criteria for pudendal neuralgia were established in the conference at Nantes and include (1) pain in the anatomical territory of the pudendal nerve, (2) pain exacerbated by sitting, (3) absence of night pain, (4) absence of objective sensory loss on clinical examination, and (5) positive anaesthetic pudendal nerve block [1].



Fig. 3 Axial image from repeat ablation with the probe positioned more proximally in the pudendal canal

Conservative therapy is an effective first line therapy for patients with pudendal neuralgia [2]. Perineural injections are employed as a complement to conservative therapy, as therapeutic interventions, and to provide diagnostic information [2, 10, 12, 24]. Surgery has been the gold standard treatment option for pudendal neuralgia. Techniques have evolved over time and the outcome literature is mixed [3, 9, 25, 26]. More recently, advanced imaging and nerve stimulator guidance have been employed in attempts to improve outcomes in patients undergoing therapeutic injections, [13–19] and neuromodulation techniques have been employed with promising initial results [23, 27, 28].

Image-guided technique

The integration of these two otherwise parallel arms (developing image guidance and ablative or modulatory techniques) was well demonstrated by Richard, et al. in their study of CT-guided pulsed radiofrequency modulation for pudendal neuralgia [29]. The study included 26 patients who underwent 53 treatments with pulsed radiofrequency in CT. Their approach, as well as the technique outlined herein, represent efforts to combine new therapeutic options with advanced imaging guidance in the setting of an established pain syndrome.

In this study, cryoablation was employed for nerve destruction. The mechanism of cryoablation-induced cell damage is well described [30], although data regarding its application for the ablation of nerves are limited [31–34]. These early clinical results suggest that transmission of sensory impulses is at least temporarily interrupted following cryoablation of the pudendal nerve, but this study is limited by its small size. Future studies may address longer term clinical effects of cryoablation on nerves in the setting of pain as well as potentially correlate the histological effects of cryoablation on myelin—as cryoablation protocols designed for destruction of neoplastic cells may or may not be applicable to neurons.

Of note, two patients did not benefit from the procedure, both with predominately perirectal pain. It may be that ablation of the pudendal nerve in the pudendal canal is distal to the sensory nerve takeoff supplying this region.

Conclusion

CT-guided percutatneous cryoablation may represent a safe and efficacious therapeutic option for selected patients with refractory pudendal neuralgia.

Conflict of interest Disclosure: We have provided paid consultation in the past for Galil Medical and specifically attended a dinner at RSNA 2012 in the capacity as paid consultants for on-label use of cryoablation.

References

1. Labat JJ, Riant T, Robert R, Amarenco G, Lefaucheur JP, Rigaud J. Diagnostic criteria for pudendal neuralgia by pudendal nerve entrapment (Nantes criteria). Neurourol Urodyn. 2008;27(4):306–10. PubMed PMID: 17828787.
2. Benson JT, Griffis K. Pudendal neuralgia, a severe pain syndrome. Am J Obstet Gynecol. 2005;192(5):1663–8. PubMed PMID: 15902174.
3. Robert R, Labat JJ, Riant T, Khalfallah M, Hamel O. Neurosurgical treatment of perineal neuralgias. Adv Tech Stand Neurosurg. 2007;32:41–59. PubMed PMID: 17907474.
4. Schraffordt SE, Tjandra JJ, Eizenberg N, Dwyer PL. Anatomy of the pudendal nerve and its terminal branches: a cadaver study. ANZ J Surg. 2004;74(1–2):23–6. PubMed PMID: 14725700.
5. Paulson JD, Baker J. De novo pudendal neuropathy after TOT-O surgery for stress urinary incontinence. JSLS J Soc Laparoendoscopic Surg Soc Laparoendoscopic Surg. 2011;15(3):326–30. PubMed PMID: 21985718. Pubmed Central PMCID: 3183551.
6. Masata J, Hubka P, Martan A. Pudendal neuralgia following transobturator inside-out tape procedure (TVT-O)–case report and anatomical study. Int Urogynecol J. 2012;23(4):505–7. PubMed PMID: 21887541.
7. Robert R, Prat-Pradal D, Labat JJ, Bensignor M, Raoul S, Rebai R, et al. Anatomic basis of chronic perineal pain: role of the pudendal nerve. Surg Radiol Anat SRA. 1998;20(2):93–8. PubMed PMID: 9658526.
8. Peltier J. Anatomical basis of transgluteal approach for pudendal neuralgia and operative technique. SRA: Surgical and radiologic anatomy; 2013. PubMed PMID: 23455364.
9. Popeney C, Ansell V, Renney K. Pudendal entrapment as an etiology of chronic perineal pain: diagnosis and treatment. Neurourol Urodyn. 2007;26(6):820–7. PubMed PMID: 17480033.
10. Filler AG. Diagnosis and treatment of pudendal nerve entrapment syndrome subtypes: imaging, injections, and minimal access surgery. Neurosurg Focus. 2009;26(2):E9. PubMed PMID: 19323602.
11. Cok OY, Eker HE, Cok T, Akin S, Aribogan A, Arslan G. Transsacral S2-S4 nerve block for vaginal pain due to pudendal neuralgia. J Minim Invasive Gynecol. 2011;18(3):401–4. PubMed PMID: 21545967.
12. Vancaillie T, Eggermont J, Armstrong G, Jarvis S, Liu J, Beg N. Response to pudendal nerve block in women with pudendal neuralgia. Pain Med. 2012;13(4):596–603. PubMed PMID: 22390343.
13. Fanucci E, Manenti G, Ursone A, Fusco N, Mylonakou I, D'Urso S, et al. Role of interventional radiology in pudendal neuralgia: a description of techniques and review of the literature. La Radiol Med. 2009;114(3):425–36. PubMed PMID: 19277838.
14. Peng PW, Tumber PS. Ultrasound-guided interventional procedures for patients with chronic pelvic pain—a description of techniques and review of literature. Pain Phys. 2008;11(2):215–24. PubMed PMID: 18354713.
15. Kim SH, Song SG, Paek OJ, Lee HJ, Park DH, Lee JK. Nerve-stimulator-guided pudendal nerve block by pararectal approach. Colorect Dis Off J Assoc Coloproctol Great Britain Ireland. 2012;14(5):611–5. PubMed PMID: 21752174.
16. Filippiadis DK, Velonakis G, Mazioti A, Alexopoulou E, Malagari A, Brountzos E. CT-guided percutaneous infiltration for the treatment of Alcock's neuralgia. Pain Phys. 2011;14(2):211–5. PubMed PMID: 21412375.
17. Carmel M, Lebel M, le Tu M. Pudendal nerve neuromodulation with neurophysiology guidance: a potential treatment option for refractory chronic pelvi-perineal pain. Int Urogynecol J. 2010;21(5):613–6. PubMed PMID: 20012596.
18. Naja MZ, Al-Tannir MA, Maaliki H, El-Rajab M, Ziade MF, Zeidan A. Nerve-stimulator-guided repeated pudendal nerve block for treatment of pudendal neuralgia. Eur J Anaesthesiol. 2006;23(5):442–4. PubMed PMID: 16573866.
19. Abdi S, Shenouda P, Patel N, Saini B, Bharat Y, Calvillo O. A novel technique for pudendal nerve block. Pain Phys. 2004;7(3):319–22. PubMed PMID: 16858468.
20. Pirro N, Sielezneff I, Le Corroller T, Ouaissi M, Sastre B, Champsaur P. Surgical anatomy of the extrapelvic part of the pudendal nerve and its applications for clinical practice. Surg Radiol Anat SRA. 2009;31(10):769–73. PubMed PMID: 19495555.
21. Antolak Jr SJ, Hough DM, Pawlina W, Spinner RJ. Anatomical basis of chronic pelvic pain syndrome: the ischial spine and pudendal nerve entrapment. Med Hypotheses. 2002;59(3):349–53. PubMed PMID: 12208168.
22. Shafik A, el-Sherif M, Youssef A, Olfat ES. Surgical anatomy of the pudendal nerve and its clinical implications. Clin Anat. 1995;8(2):110–5. PubMed PMID: 7712320.
23. Rigoard P, Delmotte A, Moles A, Hervochon R, Vrignaud T, Misbert L. Successful treatment of pudendal neuralgia with tricolumn spinal cord stimulation: case report. Neurosurgery. 2012;71(3):E757–62. discussion E63. PubMed PMID: 22653388.
24. Prat-Pradal D, Metge L, Gagnard-Landra C, Mares P, Dauzat M, Godlewski G. Anatomical basis of transgluteal pudendal nerve block. Surg Radiol Anat SRA. 2009;31(4):289–93. PubMed PMID: 19096751.
25. Hibner M, Desai N, Robertson LJ, Nour M. Pudendal neuralgia. J Minim Invasive Gynecol. 2010;17(2):148–53. PubMed PMID: 20071246.
26. Peltier J. Anatomical basis of transgluteal approach for pudendal neuralgia and operative technique. Surg Radiol Anat SRA. 2013;35(7):609–14. PubMed PMID: 23455364.
27. Marcelissen T, Van Kerrebroeck P, de Wachter S. Sacral neuromodulation as a treatment for neuropathic clitoral pain after abdominal hysterectomy. Int Urogynecol J. 2010;21(10):1305–7. PubMed PMID: 20386879, Pubmed Central PMCID: 2931644.
28. Rhame EE, Levey KA, Gharibo CG. Successful treatment of refractory pudendal neuralgia with pulsed radiofrequency. Pain Phys. 2009;12(3):633–8. PubMed PMID: 19461829.
29. Richard HMMR. CT-guided pulsed radiofrequency treatment for pudendal neuralgia. J Vasc Interv Radiol. 2014;3:S109.
30. Hartman JB, Bhojwani N, Corn DJ, Cooney MM, Haaga J, Ponsky L, et al. Incidence of hypercoagulable events after image-guided percutaneous cryoablation of renal tumors: A single-center experience. J Vasc Interven Radiol: JVIR. 2014 Mar 18. PubMed PMID: 24656177.
31. Birkenmaier C, Veihelmann A, Trouillier H, Hausdorf J, Devens C, Wegener B, et al. Percutaneous cryodenervation of lumbar facet joints: a prospective clinical trial. Int Orthop. 2007;31(4):525–30. PubMed PMID: 16927087, Pubmed Central PMCID: 2267642.
32. Campos NA, Chiles JH, Plunkett AR. Ultrasound-guided cryoablation of genitofemoral nerve for chronic inguinal pain. Pain Phys. 2009;12(6):997–1000. PubMed PMID: 19935984.
33. Koethe Y, Mannes AJ, Wood BJ. Image-guided nerve cryoablation for post-thoracotomy pain syndrome. Cardiovasc Interven Radiol. 2013 Aug 17. PubMed PMID: 23954965.
34. Niu L, Wang Y, Yao F, Wei C, Chen Y, Zhang L, et al. Alleviating visceral cancer pain in patients with pancreatic cancer using cryoablation and celiac plexus block. Cryobiology. 2013;66(2):105–11. PubMed PMID: 23267876.